

Unit 2: Computer Hardware

Short Questions

What are basic units of a Computer system?

⇒ Basic units of a computer system refer to the essential components that work together to perform input processing output and storage functions.

Define CPU and its components.

⇒ CPU, Central Processing Unit, is the 'brain' of the computer that executes instructions. It is responsible for coordinating data flow and processing tasks. It is made up of

(i) Control Unit

(ii) Arithmetic and Logical Unit

(iii) Memory Unit.

Explain the roles of buses in a computer system.

⇒ Buses are important for a computer system because buses help transfer data between components, specify memory locations for data access and manage control signals for operations.

Date _____
Page _____

What is the Jekh - execute cycle and why is it important?

⇒ Jekh Execute cycle describes how the CPU retrieves and executes instructions from memory.

This cycle is important because it is ~~the~~ a fundamental process that enables a computer's ~~oper~~ to process instructions continuously and perform complex computations.

Differentiate between main and secondary memory.

Feature	Primary Memory	Secondary Memory
Def ⁿ	Memory used by CPU to store data temporarily	Memory used for long term data storage
Speed	Very Fast	Slow compared to primary memory
Volatility	Volatile	Non-volatile

Long Question

Describe the organization of a simple computer system, including data and control paths.

⇒ A simple computer system consists of the following components

Input Unit: It accepts data and convert it into machine readable form.

CPU: Central Processing Unit is made up of 3 sub-units.

- **ALU:** Arithmetic and Logical Unit is the responsible for carrying out arithmetic and logical operations.

- **CU:** Control Unit directs the flow of data and instructions.

- **MU:** Memory Unit stores data and instruction

Output Unit: It displays the processed information in human readable form.

Storage Unit: It stores data for future use.

Date _____
Page _____

Data and Control paths also known as buses are divided into 3 types based on their purpose.

- Data Bus: It transfers data within/between components.
- Address Bus: It specifies memory location for data access.
- Control Bus: Manages control signal for operations.

Write a detailed note on Fetch Execute Cycle.

⇒ Fetch Execute Cycle describes how the CPU retrieves an instruction from memory and executes it. It generally (involves) expands to four steps.

- Fetch: The CPU retrieves an instruction from memory.
- Decode: The instruction is decoded to determine what action is required.
- Execute: The instruction is carried out by ALU and/or other unit.

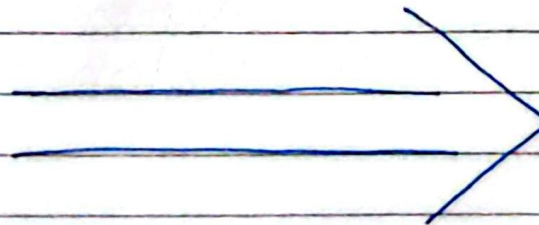
- Date _____
Page _____
- Storage: Any results are stored back into memory or used as needed.

This cycle is fundamental to the computer's operation as it enables it to process instructions continuously and perform complex calculations.

Explain the functions of Control Unit (CU) and Arithmetic and Logic Unit (ALU).

⇒ Control Unit (CU) manages and coordinates all operations on a computer. It is also responsible for interpreting instructions, generating control signals for components and coordinating (Input/Output) how I/O, memory and CPU interact.

Arithmetic and Logic Unit better known as ALU performs all arithmetic & logical operations. It also stores intermediate results in registers.



Discuss the role of peripherals in computer system with examples.

⇒ Peripherals are external devices connected to a computer that help expand its functionality.

Input devices:

Keyboard ⇒ Text input

Mouse ⇒ pointer and click

Microphone ⇒ Audio input

Touch panel ⇒ Touch input

Output devices:

Monitor ⇒ Softcopy display

Printer ⇒ Hardcopy display

Speaker ⇒ Audio output

Storage devices:

USB ⇒ Portable storage

SSD ⇒ Stationary/portable storage

Communication devices:

Network card ⇒ Connect to internet

Modem ⇒ Modulate and demodulate signal

Compare and contrast the levels of memory in a computer system.

Feature	Register	Cache	Main Memory	Secondary	Tertiary
Speed	Fastest	Very fast	Fast	Slow	slowest
Cost	Highest	Very high	Medium	Low	lowest
Capacity	Very small	Small	Moderate	Very large	Large
Location	Inside CPU	Between CPU & RAM	On Mother board	Internal / External	External